LESSON PLAN

DATE: April. 3rd

PREPARE

Objectives: By the end of the lesson the students will be able to…

1. Understand the proofs of det(AB) = (det A)(det B) and det A^T = det A;

2. Consolidate their knowledge of determinant, elementary matrices, and row picture of matrix multiplication.

Preparation (materials, resources, practice):

- Materials: lecture notes (attached), tablet;
- Practice with Leslie on Apr. 4th on Zoom.

Personal goals (what I'm working on in my teaching):

- ➢ Live writing on tablet (make fonts larger and more legible);
- ➢ Pace and tone of my lecturing;
- ➢ Time control (goal: under 30min);
- ➢ Interaction with students (ask students fill in some details of a proof).

INSTRUCT

Intro:

Bridge-in to create interest and motivation to learn:

- Recall for the class that Leslie mentioned the two theorems det(AB) = (det A)(det B) and det A^T = det A last time. We will prove the two theorems in this class.

State learning objectives and connections to course as a whole:

- Prove det(AB) = (det A)(det B) and det A^T = det A

Review:

- Definition of determinant and its properties;
- Determinant of a matrix is 0 if and only if the matrix is singular;
- One can compute the determinant of a nonsingular matrix using Gaussian elimination.

Learning Activities:

What you will do, what they will do, check for understanding (assess). Connect activities to each other. Repeat.

Activity:

Prove det(AB) = (det A)(det B)

Assess:
1. AB is nonsingular if and only if A and B are nonsingular;
2. Determinant of singular matrices is 0.

Connect:
1. *Row picture of matrix multiplication;*
2. *Defining properties of determinant.*

Activity:

Prove det A^T = det A

Assess:
1. A is nonsingular if and only if A^T is nonsingular
2. Enumeration of elementary operations

Connect:
1. *Elementary matrices*
2. *det(AB) = (det A)(det B)*

Outro:

Summarize lesson:

We have proven det(AB) = (det A)(det B) and det A^T = det A.

Make connections to rest of course or next lesson:

These two properties of determinant, like others, will be extremely useful in other proofs related to determinant and simplifying computations.

When you finish writing your plan, use this checklist to review it:

At the beginning, I plan to:

1. Use a bridge-in to capture students' interest and motivate them to learn ..... ❑
2. Make the learning outcomes clear. ..... ❑
3. Assess prior learning and student expectations ..... ❑

In the middle, I plan to:

1. Use strategies to actively involve students in the learning process ..... ❑
2. Use a variety of media to illustrate concepts and processes ..... ❑
3. Ensure that the lesson flows easily and logically ..... ❑
4. Ensure that students are learning material that is meaningful and new ..... ❑
5. Provide opportunities for practice and feedback ..... ❑
6. Review and build on related material ..... ❑

At the end, I plan to:

Provide the proper closure students find important. To do this, I will:

1. Assess what students have learned ..... ❑
2. Summarize the lesson ..... ❑
3. Connect the lesson to real life and/or the next lesson ..... ❑

Apr. 4th

Math 221/721

<u>Goals</u>

Prove: ① $\det(AB) = (\det A)(\det B)$

② $\det A^T = \det A$

<u>Last time:</u>

<u>Definition/Theorem (Determinant)</u>.

There is a unique function det, called the <u>determinant</u> that associates to each $n \times n$ matrix a real number and has the following four properties:

①. For $i \neq j$,

$$\det \begin{pmatrix} - A_1 - \\ \vdots \\ - A_i - \\ \vdots \\ - A_j - \\ \vdots \\ - A_n - \end{pmatrix} = - \det \begin{pmatrix} - A_1 - \\ \vdots \\ - A_j - \\ \vdots \\ - A_i - \\ \vdots \\ - A_n - \end{pmatrix}$$

② For scalar c,

$$\det\begin{pmatrix} -cA_1- \\ -A_2- \\ \vdots \\ -A_n- \end{pmatrix} = c\cdot\det\begin{pmatrix} -A_1- \\ -A_2- \\ \vdots \\ -A_n- \end{pmatrix}$$

③.

$$\det\begin{pmatrix} -A_1+A_1'- \\ -A_2- \\ \vdots \\ -A_n- \end{pmatrix} = \det\begin{pmatrix} -A_1- \\ -A_2- \\ \vdots \\ -A_n- \end{pmatrix} + \det\begin{pmatrix} -A_1'- \\ -A_2- \\ \vdots \\ -A_n- \end{pmatrix}$$

④ $\det I = 1.$

<u>Proposition.</u> $\det A = 0$ iff A is singular.

If A is nonsingular, $\det A$ can be computed by Gaussian Elimination.

<u>Ex.</u>

$$A = \begin{pmatrix} -1 & 3 & 5 \\ 6 & 4 & 2 \\ -2 & 5 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & -3 & -5 \\ 0 & 22 & 32 \\ 0 & -1 & -9 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & -3 & -5 \\ 0 & 22 & 32 \\ 0 & 0 & -9+\frac{32}{22} \end{pmatrix}$$

$$\det A = 1\cdot 22\cdot\left(-9+\frac{32}{22}\right) = -166.$$

<u>Proposition</u>. $\det(AB) = (\det A)(\det B)$

<u>Pf.</u> We break into two cases.

<u>Case 1</u>. B is singular.

$\Rightarrow$ AB is singular

$\Rightarrow$ $\det(AB) = \det B = 0$

$\Rightarrow$ $\det(AB) = 0 = (\det A)(\det B)$

<u>Case 2</u>. B is nonsingular.

$\Rightarrow$ $\det B \neq 0$.

Define function $d(A) = \dfrac{\det(AB)}{\det B}$.

<u>Claim</u>: $d(A) = \det A$.

Since determinant is unique w.r.t. the 4 properties above, suffices to verify that $d(A)$ satisfies these properties.

Write $A = \begin{pmatrix} A_1 \\ A_2 \\ \vdots \\ A_n \end{pmatrix}$ $B = \begin{pmatrix} B_1 \\ B_2 \\ \vdots \\ B_n \end{pmatrix}$

Then, $AB = \begin{pmatrix} A_1 B \\ A_2 B \\ \vdots \\ A_n B \end{pmatrix}$

① If $A' = \begin{pmatrix} A_1 \\ \vdots \\ A_j \\ \vdots \\ A_i \\ \vdots \\ A_n \end{pmatrix}$, $A'B = \begin{pmatrix} A_1 B \\ \vdots \\ A_j B \\ \vdots \\ A_i B \\ \vdots \\ A_n B \end{pmatrix}$

$\Rightarrow \det(A'B) = -\det(AB)$

$\Rightarrow d(A') = \dfrac{\det(A'B)}{\det B} = \dfrac{-\det(AB)}{\det B} = -d(A)$

② If $A' = \begin{pmatrix} cA_1 \\ A_2 \\ \vdots \\ A_n \end{pmatrix}$, $A'B = \begin{pmatrix} cA_1 B \\ A_2 B \\ \vdots \\ A_n B \end{pmatrix}$

$\Rightarrow \det(A'B) = c\det(AB)$

$\Rightarrow d(A') = cd(A).$

③ $A = \begin{pmatrix} A_1' + A_1'' \\ A_2 \\ \vdots \\ A_n \end{pmatrix}$, then

$$AB = \begin{pmatrix} (A_1' + A_1'')B \\ A_2B \\ \vdots \\ A_nB \end{pmatrix} = \begin{pmatrix} A_1'B + A_1''B \\ A_2B \\ \vdots \\ A_nB \end{pmatrix}$$

$$\det(AB) = \det \begin{pmatrix} A_1'B \\ A_2B \\ \vdots \\ A_nB \end{pmatrix} + \det \begin{pmatrix} A_1''B \\ A_2B \\ \vdots \\ A_nB \end{pmatrix}$$

$$= \det \begin{pmatrix} A_1' \\ A_2 \\ \vdots \\ A_n \end{pmatrix} B + \det \begin{pmatrix} A_1'' \\ A_2 \\ \vdots \\ A_n \end{pmatrix} B$$

$$\Rightarrow d(A) = d\left(\begin{pmatrix} A_1' \\ \vdots \\ A_n \end{pmatrix}\right) + d\left(\begin{pmatrix} A_1'' \\ \vdots \\ A_n \end{pmatrix}\right)$$

④. $d(I) = \dfrac{\det(IB)}{\det B} = \dfrac{\det B}{\det B} = 1$

Thus, $d(A) = \det A$. $(\det A)(\det B) = \dfrac{\det(AB)}{\det B} \det B$

$= \det(AB)$ $\square$

<u>Proposition</u> $\det A^T = \det A$.

<u>Pf.</u> Again, we break into two cases.

<u>Case 1.</u> A is singular.

$\Rightarrow$ A^T is singular

$\Rightarrow$ $\det A^T = 0 = \det A$.

<u>Case 2.</u> A is nonsingular.

We can apply Gaussian elimination to reduce A to I, i.e.,

$$E_m \cdots E_2 E_1 A = I,$$

where $E_1, \dots, E_m$ are elementary matrices.

Then,

$$A^T E_1^T E_2^T \cdots E_m^T = I$$

It follows from the proposition above that

$$(\det E_m)\cdots(\det E_2)(\det E_1)(\det A)$$
$$= \det(E_m \cdots E_2 E_1 A)$$
$$= \det(I)$$
$$= \det(A^T E_1^T E_2^T \cdots E_m^T)$$
$$= (\det A^T)(\det E_1^T)(\det E_2^T)\cdots(\det E_m^T)$$

<u>Claim:</u> For any elementary matrix E, $\det E = \det E^T$.

① <u>Swap two rows.</u>

$$E = \begin{array}{c} \\ \\ i \\ \\ j \\ \\ \end{array} \begin{array}{c} \begin{array}{cccccc} & & i & & j & \end{array} \\ \begin{pmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 0 & & 1 & \\ & & & \ddots & & \\ & & 1 & & 0 & \\ & & & & & \ddots \\ & & & & & & 1 \end{pmatrix} \end{array} \Rightarrow E = E^T$$

$$\Rightarrow \det E = \det E^T$$

② <u>Row multiplication</u>

$$E = \begin{array}{c c} & \begin{array}{ccccccc} & & i & & & \end{array} \\ i & \begin{pmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 1 & & & \\ & & m & 1 & & \\ & & & & \ddots & \\ & & & & & 1 \end{pmatrix} \end{array} \Rightarrow E = E^T$$

$$\Rightarrow \det E = \det E^T$$

③ Row addition

$$E = \begin{array}{c c} & \begin{array}{ccccccc} & j & & & & \end{array} \\ i & \begin{pmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 1 & & & \\ & & m & 1 & & \\ & & & & \ddots & \\ & & & & & 1 \end{pmatrix} \end{array}$$

$$E^T = \begin{array}{c c} & \begin{array}{ccccccc} & & i & & & \end{array} \\ j & \begin{pmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 1 & m & & \\ & & & \ddots & & \\ & & & & 1 & \\ & & & & & \ddots \\ & & & & & & 1 \end{pmatrix} \end{array}$$

Both matrices can be reduced to I by adding a multiple of a row to another.

Thus,

$$\det E = \det E^T = \det I$$

$\square$

Overall, we see that:

$$\det E_i = \det E_i^T \text{ for all } i = 1, 2, \dots, m.$$

$$\Rightarrow \det A = \det A^T \qquad \square.$$